

SIT720: Machine Learning

An Explainable Optimised Stacking Ensemble Framework for Heart Attack Prediction: Reproduction and Enhancement of a Machine Learning Approach

Student Name: Nafis Hasnat
Student ID: 226455731
Unit: SIT720 Machine Learning
Assessment: Task 11.1HD: HD Machine Learning Mini Research
Submission Date: September 2026

Abstract

Early prediction of heart disease using machine learning techniques has received significant attention due to the increasing impact of cardiovascular diseases worldwide. Ensemble learning approaches, particularly stacking-based models, have demonstrated strong predictive capability by combining multiple classifiers. However, reported high-performance results require careful validation to ensure that model effectiveness is not influenced by data quality issues or evaluation limitations.

This study reproduces a published stacking-based ensemble framework for heart attack prediction and critically analyses its performance. The original approach was implemented using six machine learning classifiers, including Logistic Regression, Decision Tree, Random Forest, XGBoost, Naive Bayes, and K-Nearest Neighbours. During the reproduction process, duplicate analysis revealed that a substantial number of repeated observations existed within the dataset, which may have contributed to overestimated performance when conventional random train-test splitting was applied.

To address this limitation, an improved explainable stacking ensemble framework was proposed. The framework incorporates duplicate removal, Random Forest-based feature importance analysis, top-feature selection, hyperparameter optimisation, and SHAP-based explainability. Experimental evaluation on the unique dataset demonstrated that the proposed framework improved prediction performance compared with the duplicate-free baseline, achieving 81.97% accuracy, 84.06% F1-score, and 0.885 AUC.

The findings highlight the importance of reliable dataset validation, model optimisation, and explainability techniques for developing trustworthy machine learning systems in healthcare prediction applications.

Keywords: Machine Learning, Heart Attack Prediction, Stacking Ensemble, Explainable Artificial Intelligence, SHAP, Feature Selection

Contents

1	Introduction	4
1.1	Research Background	4
1.2	Research Motivation	5
1.3	Research Objectives	5
2	Literature Review and Research Gap	6
2.1	Machine Learning Approaches for Heart Disease Prediction	6
2.2	Ensemble Learning Techniques	6
2.3	Explainable Artificial Intelligence in Healthcare	7
2.4	Research Gap	7
3	Reproduction of Published Machine Learning Results	9
3.1	Selected Research Paper Overview	9
3.2	Research Problem and Motivation	9
3.3	Dataset Description	10
3.4	Feature Description	10
3.5	Data Preprocessing	11
3.6	Machine Learning Models	12
3.7	Experimental Protocol	12
3.8	Evaluation Metrics	13
4	Reproduction Results and Critical Analysis	15
4.1	Individual Classifier Performance	15
4.2	Stacking Ensemble Reproduction	15
4.3	Comparison with Published Results	16
4.4	Critical Analysis of Reproduced Results	17
5	Proposed Improved Machine Learning Framework	18
5.1	Motivation and Design Objectives	18
5.2	Framework Overview	18
5.3	Duplicate-Free Dataset Preparation	20
5.4	Feature Importance Analysis	20
5.5	Feature Selection	21
5.6	Hyperparameter Optimisation	22
5.7	Optimised Stacking Ensemble	23
5.8	Explainable AI Integration	24
6	Experimental Evaluation of Proposed Framework	25
6.1	Experimental Setup	25
6.2	Performance Comparison	25
6.3	Confusion Matrix Analysis	26
6.4	ROC Curve Analysis	28

6.5	SHAP Explainability Analysis	29
7	Discussion	31
7.1	Key Findings	31
7.2	Comparison with Original Approach	31
7.3	Practical Implications	32
7.4	Limitations	33
8	Future Work	34
9	Conclusion	35
10	Project Resources	36

1 Introduction

The increasing adoption of machine learning in healthcare requires models that are not only accurate but also reliable, transparent, and reproducible. In clinical prediction tasks, achieving high classification performance alone is insufficient because incorrect evaluation may lead to misleading conclusions about model effectiveness. This research therefore investigates the relationship between dataset quality, model optimisation, and prediction reliability by reproducing an existing stacking ensemble approach and developing an improved explainable framework for heart attack prediction.

The study contributes by providing an independent evaluation of a published machine learning method, identifying potential limitations in the original experimental setup, and proposing a more robust pipeline that integrates data validation, feature optimisation, ensemble learning, and explainability.

1.1 Research Background

Heart disease remains one of the major global health challenges and is associated with significant mortality worldwide. Early identification of individuals at risk can support timely medical intervention and improve clinical decision-making. Traditional diagnosis methods mainly rely on clinical assessment and expert interpretation of patient information; however, the increasing availability of medical datasets has enabled machine learning techniques to assist healthcare professionals in identifying complex patterns associated with cardiovascular diseases.

Machine learning approaches have been widely applied in healthcare prediction tasks because of their ability to analyse multiple clinical attributes and discover relationships that may not be easily identified through conventional statistical methods. Previous studies have demonstrated that supervised learning algorithms, including Logistic Regression, Decision Trees, Random Forest, and Support Vector Machines, can provide effective solutions for heart disease prediction by using patient characteristics such as age, cholesterol level, blood pressure, chest pain type, and maximum heart rate [1, 2].

Recently, ensemble learning methods have attracted considerable attention due to their ability to combine multiple predictive models and improve classification performance. Among these approaches, stacking ensemble techniques integrate several base learners and use a meta-classifier to generate final predictions. Bhagat et al. proposed a stacking-based ensemble framework for early heart attack prediction using six machine learning classifiers, including Logistic Regression, Decision Tree, Random Forest, XGBoost, Naive Bayes, and K-Nearest Neighbours [3]. Their study reported strong predictive performance, demonstrating the potential of ensemble learning for cardiovascular risk prediction.

However, high reported predictive performance in healthcare machine learning models requires careful validation. Factors such as dataset quality, duplicated samples, feature selection strategies, and evaluation procedures can significantly influence model performance. Therefore, reproducing published machine learning studies is essential to assess their reliability and identify opportunities for improvement.

1.2 Research Motivation

The motivation of this research originates from the need to evaluate the reproducibility and reliability of high-performing machine learning approaches reported in healthcare applications. Although ensemble-based models have shown promising results for heart disease prediction, reported accuracy values may be affected by dataset characteristics and experimental settings. Therefore, validating published approaches through independent implementation is important to determine whether the reported performance can be consistently achieved.

The selected research paper by Bhagat et al. introduced an efficient stacking-based ensemble technique for early heart attack prediction and reported high predictive performance using a combination of multiple machine learning classifiers [3]. However, during the reproduction process, this study identified that the original dataset contained a large number of duplicated observations. Such data quality issues may influence model evaluation by allowing similar samples to appear in both training and testing subsets.

Therefore, this research aims not only to reproduce the existing stacking ensemble framework but also to critically analyse its limitations and develop an improved machine learning pipeline. The proposed framework focuses on improving evaluation reliability through duplicate removal, feature optimisation, hyperparameter tuning, and explainable artificial intelligence techniques.

1.3 Research Objectives

This research aims to investigate, reproduce, and improve a stacking-based machine learning framework for early heart attack prediction. The specific objectives of this study are:

1. To reproduce the stacking ensemble framework proposed in the selected research paper by implementing the reported machine learning models and evaluating their predictive performance using the original dataset.
2. To critically analyse the reproducibility of the published results by examining dataset characteristics, including duplicate observations and their potential impact on model evaluation.
3. To develop an improved machine learning framework by incorporating duplicate removal, feature importance analysis, feature selection, and hyperparameter optimisation techniques.
4. To evaluate the proposed optimised stacking ensemble model using multiple performance metrics, including accuracy, precision, recall, F1-score, and AUC.
5. To improve model transparency by applying explainable artificial intelligence techniques, particularly SHAP analysis, to identify the clinical features contributing to prediction outcomes.

2 Literature Review and Research Gap

2.1 Machine Learning Approaches for Heart Disease Prediction

Machine learning has become an important approach for healthcare prediction tasks because of its ability to identify complex relationships between clinical variables and patient outcomes. Unlike traditional statistical methods, machine learning algorithms can analyse large medical datasets and discover hidden patterns that may support early disease detection and clinical decision-making.

In heart disease prediction, supervised learning algorithms have been widely investigated using patient characteristics such as age, sex, blood pressure, cholesterol level, chest pain type, maximum heart rate, and exercise-related parameters. Mohan et al. demonstrated that hybrid machine learning techniques can effectively improve heart disease prediction performance by combining different learning strategies [1]. Similarly, Uddin et al. compared multiple supervised machine learning algorithms for disease prediction and highlighted that model performance varies depending on dataset characteristics, feature selection, and evaluation strategies [2].

Traditional classification algorithms such as Logistic Regression and Decision Trees remain popular in healthcare applications because of their simplicity and interpretability. However, advanced ensemble-based algorithms, including Random Forest and gradient boosting techniques, have demonstrated stronger capability in capturing complex patterns from medical datasets. Random Forest improves prediction stability by combining multiple decision trees and reducing variance through ensemble learning [4]. Furthermore, XGBoost provides an efficient gradient boosting framework that improves predictive performance through sequential learning and regularisation techniques [5].

Recent studies have shown that optimised machine learning and ensemble approaches can provide improved predictive performance for cardiovascular disease prediction. However, achieving high accuracy alone does not guarantee reliable healthcare applications because model performance can be affected by dataset quality, preprocessing decisions, and evaluation procedures [6, 7].

2.2 Ensemble Learning Techniques

Ensemble learning is a machine learning strategy that combines multiple individual models to produce a stronger predictive system. The main objective of ensemble methods is to reduce the limitations of individual classifiers by integrating different learning behaviours and improving generalisation performance.

Bagging, boosting, and stacking are three commonly used ensemble learning strategies. Bagging methods generate multiple models using different subsets of training data and combine their outputs to improve stability. Random Forest is a widely adopted bagging algorithm that constructs multiple decision trees and aggregates their predictions to achieve improved classification performance [4].

Boosting approaches train models sequentially, where each new learner attempts to correct errors produced by previous models. XGBoost is one of the most successful gradient boosting algorithms and has been widely applied in medical prediction tasks because of its efficiency and

strong predictive capability [5].

Among ensemble techniques, stacking provides a flexible framework by combining different types of base learners and using a meta-classifier to generate the final prediction. Unlike individual models, stacking can utilise complementary information learned by multiple classifiers, making it suitable for complex healthcare classification problems.

Bhagat et al. proposed a stacking-based ensemble framework for early heart attack prediction by combining six machine learning classifiers, including Logistic Regression, Decision Tree, Random Forest, XGBoost, Naive Bayes, and K-Nearest Neighbours, with a meta-learning strategy for final prediction [3]. Their study demonstrated the potential of stacking approaches for achieving strong predictive performance in heart disease classification.

However, ensemble models can produce misleading performance estimates when dataset quality issues, such as duplicated observations or inappropriate evaluation procedures, are not properly considered. Therefore, reliable validation and reproducibility analysis remain essential for healthcare machine learning systems.

2.3 Explainable Artificial Intelligence in Healthcare

Although machine learning models can achieve high predictive performance, many advanced algorithms operate as black-box systems where the reasoning behind model decisions is difficult to interpret. In healthcare environments, model transparency is particularly important because clinicians require understandable information regarding why a prediction has been generated.

Explainable Artificial Intelligence (XAI) aims to improve the interpretability of machine learning models by identifying the contribution of individual features to prediction outcomes. Among different XAI techniques, SHAP (SHapley Additive exPlanations) has gained significant attention because it provides both global and local explanations of machine learning predictions. Lundberg and Lee introduced SHAP as a unified framework for interpreting model predictions using feature contribution values based on cooperative game theory [8].

SHAP analysis allows researchers to determine which clinical variables have the largest influence on prediction outcomes. In heart disease prediction, explainable methods can provide insights into the importance of factors such as chest pain characteristics, maximum heart rate, cholesterol level, age, and exercise-related measurements.

The importance of interpretable machine learning has increased in healthcare because prediction accuracy alone is insufficient for clinical decision support. Models should provide reliable explanations alongside predictions to improve trust and practical usability [9]. Recent research has also demonstrated that tree-based explainability methods can provide meaningful global and individual-level interpretations for complex machine learning models [10].

Therefore, this research integrates SHAP-based analysis into the proposed stacking ensemble framework to evaluate feature contributions and improve the transparency of heart attack prediction.

2.4 Research Gap

Existing studies have demonstrated that machine learning and ensemble learning approaches can achieve promising performance for heart disease prediction. However, several limitations

remain regarding the reliability, reproducibility, and interpretability of these models.

The selected research paper by Bhagat et al. reported high predictive performance using a stacking-based ensemble technique for early heart attack prediction [3]. However, independent reproduction of machine learning studies is necessary because reported results may be influenced by dataset characteristics, preprocessing strategies, feature selection methods, and evaluation procedures.

Furthermore, many studies focus primarily on improving predictive accuracy while providing limited investigation into data quality issues. Duplicate observations within datasets can introduce bias and may result in overestimated performance when similar samples appear in both training and testing subsets. Therefore, careful dataset validation is required before evaluating healthcare prediction models.

Another important limitation is the lack of interpretability in many high-performing ensemble models. Although these models may achieve strong classification results, healthcare applications require transparent decision-making processes that allow users to understand the contribution of clinical features.

Based on these limitations, this research identifies three major gaps. First, published stacking-based heart disease prediction approaches require independent reproduction and critical evaluation. Second, the effect of duplicate observations on model performance requires further investigation. Third, improved frameworks should integrate optimisation strategies and explainability techniques to achieve both reliable performance and transparent predictions.

To address these gaps, this study reproduces the existing stacking ensemble framework, evaluates performance using a duplicate-free dataset, and proposes an improved explainable stacking approach incorporating feature selection, hyperparameter optimisation, and SHAP-based interpretation.

3 Reproduction of Published Machine Learning Results

3.1 Selected Research Paper Overview

The selected research paper reproduced in this study is titled “An efficient stacking-based ensemble technique for early heart attack prediction” by Bhagat et al. [3]. The study proposed a stacking-based ensemble learning framework for early heart attack prediction by combining multiple machine learning classifiers to improve predictive performance.

The original framework utilised six individual machine learning classifiers as base learners, including Logistic Regression, Decision Tree, Random Forest, XGBoost, Naive Bayes, and K-Nearest Neighbours. A meta-classifier was then used to combine the outputs of these base models and generate the final prediction. The authors reported that the stacking approach achieved strong classification performance compared with individual machine learning algorithms.

The main objective of the original study was to develop an effective machine learning framework capable of predicting heart attack risk using clinical attributes. The research demonstrated the potential of ensemble learning methods for cardiovascular disease prediction by integrating different classification techniques into a single predictive system.

In this research, the proposed stacking framework was independently reproduced using the dataset provided with the task. The reproduction process followed the general methodology described in the original paper, including dataset preprocessing, model implementation, training, and evaluation using standard classification performance metrics.

3.2 Research Problem and Motivation

The primary research problem addressed in the selected study is the development of an effective machine learning framework for early heart attack prediction. Accurate identification of individuals at risk can support early intervention and assist healthcare professionals in making informed decisions. The original study focused on improving prediction performance by combining multiple machine learning classifiers through a stacking ensemble approach [3].

Although the original research reported strong predictive performance, machine learning results in healthcare applications require careful validation before being considered reliable. Model performance can be influenced by several factors, including dataset composition, preprocessing techniques, feature selection strategies, and evaluation procedures. Therefore, reproducing published machine learning approaches is important for determining whether reported results can be independently achieved under similar experimental conditions.

During the reproduction process, this study identified a significant data quality issue within the provided heart disease dataset. A large proportion of observations were found to be duplicated, which raised concerns regarding the reliability of random train-test evaluation. When similar records are distributed between the training and testing sets, machine learning models may achieve artificially high performance because they are evaluated on observations with highly similar patterns.

Therefore, the motivation of this reproduction study extends beyond simply replicating the original results. It aims to critically examine the reliability of the reported performance, investigate the impact of duplicated observations, and develop an improved evaluation frame-

work. The study further proposes an enhanced stacking approach by incorporating duplicate removal, feature selection, hyperparameter optimisation, and explainable artificial intelligence techniques.

3.3 Dataset Description

The dataset used in this research is a heart disease dataset containing clinical information collected from patients for binary classification of heart disease presence. The dataset consists of patient demographic information, clinical measurements, and diagnostic indicators. The target variable represents the presence or absence of heart disease, where the classification task aims to predict whether a patient belongs to the positive or negative heart disease class.

The original dataset contained 1,025 patient records with 13 predictive features and one target variable. The input variables represent commonly used clinical indicators, including demographic characteristics, blood pressure measurements, cholesterol levels, exercise-related information, and cardiac diagnostic attributes.

During the initial data investigation, duplicate analysis was performed to examine the reliability of the dataset. The analysis identified 723 duplicated records, representing approximately 70.54% of the complete dataset. After removing duplicate observations, the dataset was reduced from 1,025 records to 302 unique patient records. This duplicate-free dataset was subsequently used for additional experiments to provide a more realistic evaluation of model generalisation performance.

Table 1 presents a summary of the dataset characteristics used in this study.

Table 1: Dataset Summary

Dataset Property	Value
Original number of records	1025
Number of predictive features	13
Target variable	Heart disease status
Duplicate records identified	723
Duplicate percentage	70.54%
Unique records after duplicate removal	302
Classification type	Binary classification

3.4 Feature Description

The dataset contains 13 clinical input features used to predict the presence of heart disease. These variables represent demographic information, physiological measurements, and diagnostic indicators commonly associated with cardiovascular risk assessment. Table 2 presents the description and data type of each feature used in this study.

Table 2: Description of Dataset Features

Feature	Description	Data Type
age	Patient age	Numerical
sex	Patient gender indicator	Categorical
cp	Chest pain type	Categorical
trestbps	Resting blood pressure	Numerical
chol	Serum cholesterol level	Numerical
fbs	Fasting blood sugar indicator	Categorical
restecg	Resting electrocardiographic results	Categorical
thalach	Maximum heart rate achieved	Numerical
exang	Exercise-induced angina	Categorical
oldpeak	ST depression induced by exercise	Numerical
slope	Slope of peak exercise ST segment	Categorical
ca	Number of major vessels coloured by fluoroscopy	Categorical
thal	Thalassemia-related diagnostic indicator	Categorical

The features consist of both numerical and categorical variables. Numerical attributes include age, resting blood pressure, cholesterol level, maximum heart rate, and ST depression values, while categorical attributes represent clinical classification indicators such as chest pain type, exercise-induced angina, and thalassemia status. Appropriate preprocessing was therefore applied before model training to ensure that all variables were suitable for machine learning algorithms.

3.5 Data Preprocessing

Data preprocessing is an essential step in machine learning workflows because the quality and representation of input data directly influence model performance. In this study, preprocessing procedures were applied to prepare the heart disease dataset for classification experiments and ensure compatibility with different machine learning algorithms.

Initially, the dataset was separated into input features and the target variable. The 13 clinical attributes were considered as independent variables, while the heart disease status was used as the binary target variable. The dataset was then examined for data quality issues, including duplicate observations.

Duplicate analysis was performed before model development to evaluate the reliability of the dataset. The analysis identified 723 duplicated records within the original 1,025 observations. For the initial reproduction experiment, the complete dataset was retained to replicate the original research setting. For further evaluation, duplicate records were removed, resulting in a unique dataset containing 302 patient observations.

The dataset contained both numerical and categorical features. Numerical features, including age, resting blood pressure, cholesterol, maximum heart rate, and oldpeak, were processed using feature scaling to standardise their ranges. Categorical features, including chest pain type, fasting blood sugar, resting ECG, exercise-induced angina, slope, number of vessels, and thalassemia indicators, were handled using appropriate preprocessing transformations.

A preprocessing pipeline was implemented to ensure consistent data transformation during

model training and testing. The processed feature set was then divided into training and testing subsets using an 80:20 split strategy. Stratified splitting was applied to preserve the distribution of the target classes across both subsets. The same preprocessing procedure was applied consistently across all machine learning experiments to provide a fair comparison between different models.

3.6 Machine Learning Models

To reproduce the original stacking-based heart attack prediction framework, six supervised machine learning classifiers were implemented as base learners. These models were selected because they represent different learning strategies, including linear classification, tree-based learning, probabilistic modelling, and distance-based classification. The combination of diverse classifiers allows the stacking ensemble to utilise complementary patterns learned from the clinical features.

Logistic Regression (LR): Logistic Regression is a widely used linear classification algorithm for binary prediction problems. It estimates the probability of a target class based on a weighted combination of input variables and provides a simple and interpretable baseline model for healthcare prediction tasks.

Decision Tree (DT): Decision Tree classifiers perform prediction by recursively splitting data based on feature conditions. They can capture non-linear relationships between clinical variables and provide relatively interpretable decision rules.

Random Forest (RF): Random Forest is an ensemble learning algorithm that combines multiple decision trees to improve prediction stability and reduce overfitting. Each tree is trained using different subsets of data and features, and the final prediction is generated through aggregation of individual tree outputs [4].

Extreme Gradient Boosting (XGBoost): XGBoost is a gradient boosting algorithm that builds decision trees sequentially, where each new tree attempts to correct the errors of previous learners. The algorithm provides strong predictive capability through regularisation and efficient optimisation techniques [5].

Naive Bayes (NB): Naive Bayes is a probabilistic classification algorithm based on Bayes' theorem. It assumes conditional independence between input features and calculates the probability of each class based on observed feature distributions.

K-Nearest Neighbours (KNN): K-Nearest Neighbours is a distance-based classification algorithm that predicts the class of a new sample by analysing the labels of neighbouring observations. The method can capture local patterns within the feature space.

The outputs of these individual classifiers were further combined using a stacking ensemble approach. In the reproduced framework, the base learners generated predictions that were used by a meta-classifier to produce the final heart disease classification outcome. This approach follows the methodology reported by Bhagat et al. [3].

3.7 Experimental Protocol

The experimental protocol was designed to reproduce the original stacking-based heart attack prediction framework and evaluate its reliability under different dataset conditions. All ex-

periments were implemented using Python-based machine learning libraries, with consistent preprocessing and evaluation procedures applied across different models.

The reproduction process was divided into three main experimental stages. In the first stage, the original dataset containing 1,025 records was used to reproduce the performance reported in the selected research paper. The six individual classifiers were trained and evaluated, followed by the implementation of the stacking ensemble model. This experiment aimed to verify whether the reported performance could be independently reproduced.

In the second stage, duplicate observations identified during dataset analysis were removed to create a duplicate-free dataset containing 302 unique records. The same machine learning models and evaluation procedure were applied to this cleaned dataset. This experiment was conducted to investigate the influence of duplicated observations on model performance and provide a more realistic assessment of generalisation capability.

In the third stage, an improved machine learning framework was developed using the duplicate-free dataset. The proposed framework incorporated Random Forest-based feature importance analysis, selection of the most influential clinical features, hyperparameter optimisation of selected base learners, and an optimised stacking ensemble model. SHAP-based explainability analysis was also performed to identify the contribution of individual features to prediction outcomes.

For all experiments, the dataset was divided into training and testing subsets using an 80:20 stratified split. Model performance was evaluated on the unseen testing data to ensure a consistent comparison between different approaches. This experimental design enables both reproduction of the existing method and critical evaluation of the proposed improvements.

3.8 Evaluation Metrics

The performance of the machine learning models was evaluated using multiple classification metrics to provide a comprehensive assessment of predictive performance. Since healthcare prediction tasks require consideration of both correct predictions and classification errors, accuracy alone is insufficient for evaluating model effectiveness. Therefore, accuracy, precision, recall, F1-score, and area under the receiver operating characteristic curve (AUC) were used in this study.

Accuracy: Accuracy measures the proportion of correctly classified samples among all testing observations. It provides an overall indication of model prediction performance and is calculated as:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad (1)$$

where TP , TN , FP , and FN represent true positives, true negatives, false positives, and false negatives, respectively.

Precision: Precision represents the proportion of correctly predicted positive cases among all samples predicted as positive. It is particularly important in medical applications because false positive predictions may lead to unnecessary clinical investigations.

$$Precision = \frac{TP}{TP + FP} \quad (2)$$

Recall: Recall measures the ability of a model to correctly identify actual positive heart disease cases. A high recall value is important in healthcare applications because failing to identify patients at risk may have serious consequences.

$$Recall = \frac{TP}{TP + FN} \quad (3)$$

F1-score: The F1-score provides a balanced measure between precision and recall by calculating their harmonic mean.

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall} \quad (4)$$

Area Under the ROC Curve (AUC): The AUC measures the ability of a classifier to distinguish between positive and negative classes across different classification thresholds. A higher AUC value indicates stronger discrimination capability between heart disease and non-heart disease cases.

These evaluation metrics were applied consistently across all experiments, including the reproduced stacking model, duplicate-free baseline models, and the proposed optimised stacking framework. This multi-metric evaluation provides a more reliable comparison of model performance and classification behaviour.

4 Reproduction Results and Critical Analysis

4.1 Individual Classifier Performance

Before implementing the stacking ensemble model, the performance of individual machine learning classifiers was evaluated using the original dataset containing 1,025 records. The objective of this experiment was to analyse the predictive capability of each base learner and compare their individual performance with the combined ensemble approach.

Six machine learning algorithms were evaluated, including Logistic Regression, Decision Tree, Random Forest, XGBoost, Naive Bayes, and K-Nearest Neighbours. Each model was trained using the same preprocessing pipeline and evaluated on the testing dataset using accuracy, precision, recall, F1-score, and AUC metrics.

Table 3 presents the performance comparison of the individual classifiers.

Table 3: Performance of Individual Machine Learning Models on Original Dataset

Model	Accuracy	Precision	Recall	F1-score	AUC
Logistic Regression	80.98	76.19	91.43	83.12	92.98
Decision Tree	98.54	100.00	97.14	98.55	98.57
Random Forest	100.00	100.00	100.00	100.00	100.00
XGBoost	100.00	100.00	100.00	100.00	100.00
Naive Bayes	82.93	80.70	87.62	84.02	90.43
KNN	86.34	87.38	85.71	86.54	96.29

The results demonstrate that tree-based ensemble models achieved the strongest performance on the original dataset. Random Forest and XGBoost both achieved perfect classification results across all evaluation metrics. Although these results indicate strong predictive capability, such extremely high performance requires further investigation because healthcare datasets with limited sample sizes may be sensitive to data characteristics and evaluation procedures.

The individual classifier results provide the foundation for the subsequent stacking ensemble experiment, where predictions from multiple models are combined to evaluate whether ensemble learning can further improve predictive performance.

4.2 Stacking Ensemble Reproduction

After evaluating the individual classifiers, a stacking ensemble model was implemented to reproduce the methodology proposed in the selected research paper. Stacking combines predictions from multiple base learners and uses a higher-level meta-model to generate the final classification output. This approach aims to utilise the complementary strengths of different machine learning algorithms and improve overall predictive performance.

In this study, the six previously evaluated classifiers, including Logistic Regression, Decision Tree, Random Forest, XGBoost, Naive Bayes, and K-Nearest Neighbours, were used as base learners. The predictions generated by these models were combined and provided as input to the stacking classifier to produce the final heart disease prediction.

The stacking ensemble was first evaluated using the original dataset containing 1,025 records, following the experimental setup used during reproduction. The model was trained and tested

using the same preprocessing pipeline and evaluation strategy applied to the individual classifiers.

Table 4 presents the performance of the reproduced stacking ensemble model.

Table 4: Reproduced Stacking Ensemble Performance on Original Dataset

Model	Accuracy	Precision	Recall	F1-score	AUC
Stacking Ensemble	100.00	100.00	100.00	100.00	100.00

The reproduced stacking model achieved perfect performance across all evaluation metrics on the original dataset. These results were consistent with the strong performance reported in the original study, where the authors demonstrated the effectiveness of stacking-based ensemble learning for early heart attack prediction [3].

Although the reproduced results indicate excellent predictive performance, such results require further investigation because perfect classification on a medical dataset with a limited number of observations may indicate possible sensitivity to dataset characteristics. Therefore, additional experiments were conducted using a duplicate-free dataset to evaluate whether the performance remained consistent under a more reliable evaluation setting.

4.3 Comparison with Published Results

The reproduced stacking ensemble model was compared with the performance reported in the original research paper to evaluate the consistency of the implementation. The original study reported an accuracy of 98.53% for the proposed stacking framework, demonstrating the effectiveness of combining multiple machine learning classifiers for heart attack prediction [3].

In this research, the stacking ensemble framework was independently implemented using the same dataset characteristics and similar machine learning models. The reproduced model achieved an accuracy of 100.00%, with precision, recall, F1-score, and AUC values also reaching 100.00%. Table 5 presents the comparison between the published result and the reproduced performance.

Table 5: Comparison Between Published and Reproduced Stacking Results

Method	Accuracy	Precision	Recall	F1-score	AUC
Original Paper Stacking	98.53	–	–	–	–
Reproduced Stacking Ensemble	100.00	100.00	100.00	100.00	100.00

The reproduced results demonstrate that the general methodology of the original stacking framework can be successfully implemented. However, the slightly higher performance achieved during reproduction also highlights the importance of investigating dataset characteristics and experimental conditions. In healthcare machine learning applications, extremely high predictive performance should be carefully examined because factors such as duplicated observations, data leakage, or evaluation design may influence the measured results.

Therefore, a further analysis was conducted by removing duplicate observations from the dataset and evaluating the stacking model on unique patient records. This additional experiment

aims to determine whether the reported performance remains consistent when evaluated using a more reliable dataset representation.

4.4 Critical Analysis of Reproduced Results

The reproduction experiment successfully achieved performance comparable to the original study, with the stacking ensemble model obtaining 100% accuracy on the original dataset. Although these results indicate strong predictive capability, the unusually high performance requires further investigation, particularly in the context of healthcare datasets where reliable generalisation is essential.

During dataset quality analysis, a substantial number of duplicated observations were identified. The original dataset contained 1,025 records, of which 723 records were found to be duplicated. This represents approximately 70.54% of the complete dataset. After removing duplicated observations, the dataset was reduced to 302 unique patient records.

The presence of duplicated samples can influence machine learning evaluation because random train-test splitting may distribute identical or highly similar observations across both training and testing subsets. As a result, the model may perform exceptionally well by recognising patterns from previously observed samples rather than learning generalisable relationships. This issue is particularly important in healthcare applications because overestimated model performance may lead to unrealistic expectations regarding clinical deployment.

To investigate the impact of duplicated observations, the stacking ensemble model was re-trained and evaluated using the duplicate-free dataset. The performance decreased substantially compared with the original experiment, indicating that the initial perfect classification performance was strongly influenced by the dataset composition.

The duplicate analysis highlights the importance of performing comprehensive data validation before developing machine learning models for medical prediction tasks. While the reproduced results confirm that the original methodology can be implemented successfully, the additional evaluation demonstrates that high performance metrics should be interpreted carefully when dataset quality issues are present.

Therefore, this study extends beyond simple reproduction by critically examining the reliability of the reported results and establishing a more robust evaluation framework for heart attack prediction.

5 Proposed Improved Machine Learning Framework

5.1 Motivation and Design Objectives

The reproduction and critical analysis of the original stacking ensemble framework revealed that achieving high predictive performance alone is insufficient for developing reliable health-care machine learning systems. Although the reproduced model achieved perfect classification performance on the original dataset, further investigation identified that duplicated observations significantly affected model evaluation. This highlights the importance of improving data validation, optimisation strategies, and model interpretability.

The proposed framework aims to address the limitations identified during the reproduction process by developing a more robust and explainable stacking ensemble approach for heart attack prediction. Instead of directly optimising performance on the original dataset, the proposed approach focuses on improving the reliability and generalisation capability of the model using a duplicate-free dataset.

The main objectives of the proposed framework are:

1. To remove duplicated observations and evaluate model performance using a more reliable representation of patient data.
2. To identify the most influential clinical features using feature importance analysis and reduce unnecessary variables.
3. To optimise selected machine learning models through hyperparameter tuning to improve predictive performance.
4. To develop an optimised stacking ensemble model by combining multiple diverse classifiers.
5. To incorporate explainable artificial intelligence techniques to analyse feature contributions and improve model transparency.

The proposed framework integrates data validation, feature optimisation, ensemble learning, and explainability into a single pipeline. This design aims to provide a more reliable and interpretable solution for heart attack prediction compared with the directly reproduced approach.

5.2 Framework Overview

The proposed framework introduces a systematic pipeline that integrates data validation, feature optimisation, ensemble learning, and explainable artificial intelligence. The objective of this framework is to overcome the limitations identified during the reproduction experiment while maintaining strong predictive performance on a more reliable dataset.

The proposed workflow consists of several sequential stages. First, the original heart disease dataset is analysed to identify and remove duplicated observations. This step reduces the risk of biased evaluation caused by identical samples being present in both training and testing subsets.

Following data cleaning, feature importance analysis is performed using a tree-based model to identify the clinical variables that contribute most significantly to prediction outcomes. The

most influential features are selected to reduce unnecessary complexity and improve model efficiency.

The selected feature subset is then used for hyperparameter optimisation of individual machine learning classifiers. Optimised base learners are combined through a stacking ensemble architecture, where multiple classifiers contribute different learning patterns to the final prediction.

Finally, explainable artificial intelligence is integrated into the framework using SHAP analysis. This provides insight into the contribution of individual clinical features and improves the transparency of the final prediction model.

Figure 1 illustrates the overall architecture of the proposed optimised stacking ensemble framework.

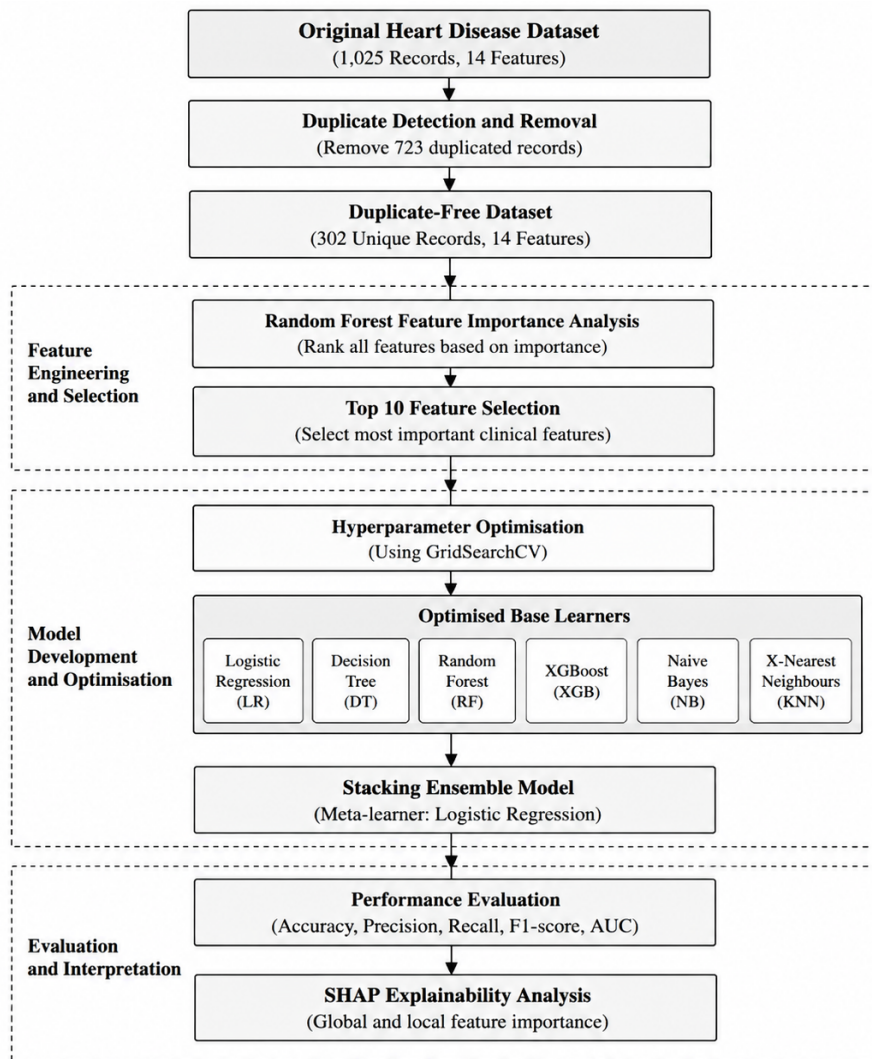


Figure 1: Overview of the proposed optimised stacking ensemble framework.

Figure 1: Overview of the proposed optimised stacking ensemble framework.

5.3 Duplicate-Free Dataset Preparation

A major limitation identified during the reproduction experiment was the presence of a large number of duplicated observations within the original dataset. Although the original stacking framework achieved excellent predictive performance, the presence of repeated records may affect the reliability of model evaluation by allowing similar patient samples to appear in both training and testing subsets.

To address this issue, duplicate analysis was performed before developing the proposed framework. The original dataset contained 1,025 records, and the analysis identified 723 duplicated observations. These duplicated records represented approximately 70.54% of the complete dataset. After removing duplicate samples, the dataset was reduced to 302 unique patient records.

The duplicate-free dataset was used as the foundation for all subsequent experiments in the proposed framework. This approach provides a more realistic evaluation environment by reducing the possibility that the model learns from repeated patient patterns rather than discovering generalisable relationships between clinical features and heart disease outcomes.

Table 6 presents the comparison between the original dataset and the duplicate-free dataset.

Table 6: Impact of Duplicate Removal on Dataset Size

Dataset Condition	Number of Records	Change
Original dataset	1025	—
Duplicate records identified	723	70.54%
Unique records after removal	302	-70.54%

The removal of duplicated observations represents an important improvement over the reproduced approach because it focuses on evaluating model performance using unique patient information. This ensures that subsequent optimisation and ensemble learning experiments are based on a more reliable dataset structure.

5.4 Feature Importance Analysis

Feature selection is an important step in developing efficient machine learning models because irrelevant or less informative variables may increase model complexity and reduce generalisation capability. In the proposed framework, Random Forest feature importance analysis was applied to identify the clinical variables that contributed most significantly to heart attack prediction.

Random Forest was selected for feature importance analysis because tree-based ensemble models can estimate the contribution of individual features by measuring how effectively each variable improves classification decisions across multiple decision trees [4]. The importance scores obtained from the trained model were used to rank the 13 input features according to their predictive contribution.

The feature importance analysis identified chest pain type (cp), maximum heart rate achieved (thalach), thalassemia indicator (thal), number of major vessels (ca), and ST depression value (oldpeak) as the most influential predictors. These features showed higher contribution scores

compared with other clinical variables, indicating their stronger relationship with heart disease classification.

Figure 2 presents the feature importance ranking generated from the Random Forest model.

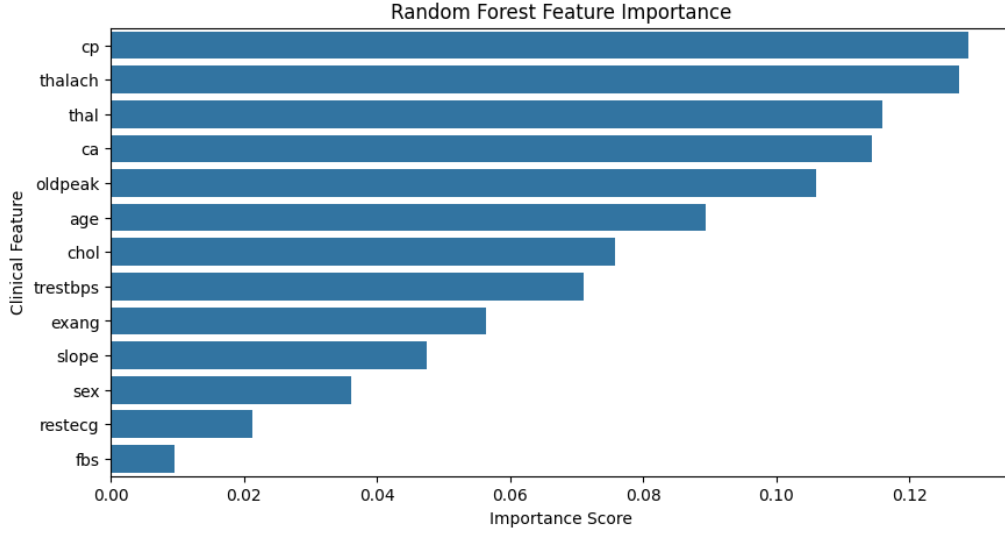


Figure 2: Feature importance ranking obtained from Random Forest analysis.

The feature importance analysis provided a data-driven approach for reducing the number of input variables while retaining the most informative clinical characteristics. Based on the ranking results, the top 10 features were selected for the proposed optimised stacking framework. This reduction aimed to improve model efficiency while maintaining predictive capability.

5.5 Feature Selection

Based on the feature importance analysis, the most influential clinical variables were selected to construct a more efficient predictive model. Feature selection was introduced in the proposed framework to reduce unnecessary complexity while maintaining the information required for accurate heart attack prediction.

The original dataset contained 13 input features. Although all features provide potential clinical information, the importance analysis demonstrated that some variables contributed more strongly to the classification outcome than others. Therefore, the top-ranked features were selected according to their importance scores generated from the Random Forest model.

The selected feature subset consisted of the following 10 clinical variables:

- Chest pain type (*cp*)
- Maximum heart rate achieved (*thalach*)
- Thalassemia indicator (*thal*)
- Number of major vessels (*ca*)
- ST depression induced by exercise (*oldpeak*)

- Age (*age*)
- Cholesterol level (*chol*)
- Resting blood pressure (*trestbps*)
- Exercise-induced angina (*exang*)
- Slope of peak exercise ST segment (*slope*)

The selected features represent important demographic, physiological, and diagnostic characteristics associated with cardiovascular risk. Features with lower importance scores were excluded from the proposed model to reduce feature redundancy and improve computational efficiency.

The reduced feature set was then used for subsequent hyperparameter optimisation and stacking ensemble training. This feature selection strategy allows the proposed framework to focus on the most informative clinical variables while maintaining a balance between predictive performance and model simplicity.

5.6 Hyperparameter Optimisation

Hyperparameter optimisation was introduced in the proposed framework to improve the performance of individual base learners before combining them through the stacking ensemble architecture. Default model parameters may not always provide optimal performance for a specific dataset; therefore, systematic parameter search was performed to identify suitable configurations.

In this study, Grid Search Cross-Validation was applied to selected machine learning algorithms. Grid Search evaluates different combinations of predefined hyperparameter values and selects the configuration that achieves the highest cross-validation performance. This approach helps reduce manual parameter selection and provides a more objective optimisation process.

Hyperparameter optimisation was performed for Random Forest, XGBoost, and Logistic Regression models because these algorithms represent important components of the final ensemble framework.

For the Random Forest classifier, the optimal configuration identified through Grid Search was:

- Number of estimators: 200
- Maximum tree depth: 3
- Minimum samples split: 2

For the XGBoost classifier, the selected optimal parameters were:

- Learning rate: 0.1
- Maximum tree depth: 3

- Number of estimators: 50

For Logistic Regression, the optimised parameters were:

- Regularisation parameter (C): 0.01
- Solver: lbfgs

Table 7 summarises the optimised parameter settings used in the proposed framework.

Table 7: Optimised Hyperparameter Configurations

Model	Parameter	Optimal Value
Random Forest	n_estimators	200
Random Forest	max_depth	3
Random Forest	min_samples_split	2
XGBoost	learning_rate	0.1
XGBoost	max_depth	3
XGBoost	n_estimators	50
Logistic Regression	C	0.01
Logistic Regression	solver	lbfgs

The optimised parameters were then used to construct the final stacking ensemble model. By combining tuned base learners rather than default models, the proposed framework aims to achieve improved predictive performance while maintaining better generalisation capability on the duplicate-free dataset.

5.7 Optimised Stacking Ensemble

The final component of the proposed framework is the optimised stacking ensemble model, which combines multiple tuned machine learning classifiers to improve prediction reliability. Unlike the reproduced stacking model, which was evaluated using the original dataset containing duplicated observations, the proposed ensemble was trained and evaluated using the duplicate-free dataset with the selected 10 clinical features.

The optimised stacking framework consists of six diverse base learners: Logistic Regression, Decision Tree, Random Forest, XGBoost, Naive Bayes, and K-Nearest Neighbours. Among these models, Random Forest, XGBoost, and Logistic Regression were optimised using Grid Search Cross-Validation, while the remaining classifiers were included to provide additional diversity within the ensemble.

The predictions generated by the optimised base learners were combined using a stacking architecture, where a meta-classifier learned from the outputs of the individual models and generated the final heart attack prediction. This approach allows the ensemble to utilise different learning patterns captured by linear, tree-based, probabilistic, and distance-based classifiers.

The optimised stacking ensemble was evaluated using the duplicate-free dataset containing 302 unique patient records. The model achieved an accuracy of 81.97%, precision of 80.56%, recall of 87.88%, F1-score of 84.06%, and AUC of 88.53%. The complete performance results are presented in Table 8.

Table 8: Performance of Proposed Optimised Stacking Ensemble

Evaluation Metric	Performance (%)
Accuracy	81.97
Precision	80.56
Recall	87.88
F1-score	84.06
AUC	88.53

The proposed model achieved improved performance compared with the duplicate-free baseline stacking model, which obtained an accuracy of 75.41%. The improvement demonstrates that feature selection and hyperparameter optimisation contributed to better classification capability on the cleaned dataset.

Although the proposed framework did not achieve the artificially high performance observed on the original duplicated dataset, it provides a more realistic evaluation of model behaviour by prioritising data reliability, optimisation, and interpretability. This makes the proposed approach more suitable for developing trustworthy machine learning systems for healthcare prediction.

5.8 Explainable AI Integration

Although the proposed optimised stacking ensemble achieved improved predictive performance, understanding the reasons behind model predictions is essential for healthcare applications. Therefore, explainable artificial intelligence was incorporated into the proposed framework to improve model transparency.

SHAP (SHapley Additive exPlanations) was applied to interpret the behaviour of the optimised model. SHAP provides a unified approach for explaining machine learning predictions by estimating the contribution of each feature to the final prediction output [8]. This enables both global interpretation of feature importance and analysis of individual prediction decisions.

The integration of SHAP allows the proposed framework to provide not only classification results but also information regarding the contribution of clinical features to model decisions. This improves the interpretability of the stacking ensemble and supports the development of more transparent healthcare machine learning systems.

6 Experimental Evaluation of Proposed Framework

6.1 Experimental Setup

The proposed optimised stacking ensemble framework was evaluated using the duplicate-free heart disease dataset to measure its predictive capability under a more reliable experimental setting. Unlike the reproduction experiment, which used the original dataset containing duplicated observations, this evaluation was performed using only unique patient records.

The duplicate removal process reduced the dataset from 1,025 records to 302 unique observations. The selected feature subset obtained from Random Forest feature importance analysis was used for model training. The final feature set consisted of 10 clinical variables: chest pain type (*cp*), maximum heart rate achieved (*thalach*), thalassemia indicator (*thal*), number of major vessels (*ca*), oldpeak, age, cholesterol level (*chol*), resting blood pressure (*trestbps*), exercise-induced angina (*exang*), and slope.

The dataset was divided into training and testing subsets using an 80:20 stratified split. This resulted in 241 training samples and 61 testing samples. Feature scaling was applied to standardise numerical variables before training the machine learning models.

The experimental evaluation included six optimised and base machine learning classifiers: Logistic Regression, Decision Tree, Random Forest, XGBoost, Naive Bayes, and K-Nearest Neighbours. These classifiers were integrated into a stacking ensemble architecture, where a meta-classifier combined the predictions from individual learners to generate the final classification output.

The proposed framework was evaluated using multiple performance metrics, including accuracy, precision, recall, F1-score, and area under the receiver operating characteristic curve (AUC). These metrics were selected to provide a comprehensive evaluation of classification performance, particularly considering the importance of correctly identifying positive heart disease cases in healthcare applications.

6.2 Performance Comparison

The performance of the proposed optimised stacking ensemble was compared with the duplicate-free baseline stacking model to evaluate the contribution of feature selection and hyperparameter optimisation. Both models were evaluated using the same duplicate-free dataset containing 302 unique patient records and the same testing procedure.

The baseline stacking model represents the reproduced ensemble framework evaluated after removing duplicated observations, while the proposed model incorporates additional optimisation strategies, including Random Forest-based feature selection and hyperparameter tuning of selected base learners.

Table 9 presents the performance comparison between the two approaches.

Table 9: Performance Comparison Between Baseline and Proposed Framework

Model	Accuracy	Precision	Recall	F1-score	AUC
Duplicate-Free Stacking Ensemble	75.41	76.47	78.79	77.61	86.80
Proposed Optimised Stacking Ensemble	81.97	80.56	87.88	84.06	88.53

The proposed framework achieved improved performance across all major evaluation metrics compared with the duplicate-free baseline model. Accuracy increased from 75.41% to 81.97%, representing a 6.56 percentage point improvement. The F1-score also improved from 77.61% to 84.06%, indicating a better balance between precision and recall.

The most notable improvement was observed in recall, which increased from 78.79% to 87.88%. This improvement is particularly relevant for heart disease prediction because identifying patients with potential disease risk is an important objective in healthcare screening applications.

The AUC value increased from 86.80% to 88.53%, demonstrating improved discrimination capability between positive and negative classes. These results suggest that feature selection and hyperparameter optimisation improved the classification ability of the stacking ensemble while maintaining evaluation on a more reliable duplicate-free dataset.

Overall, the performance comparison demonstrates that the proposed framework provides a more robust solution than the unoptimised duplicate-free baseline by combining improved feature representation, tuned classifiers, and ensemble learning.

6.3 Confusion Matrix Analysis

Confusion matrix analysis was performed to further evaluate the classification behaviour of the proposed optimised stacking ensemble. Unlike aggregate metrics, the confusion matrix provides detailed information about correct predictions and classification errors by showing the number of true positives, true negatives, false positives, and false negatives.

The duplicate-free baseline stacking model produced the confusion matrix shown in Figure 3. The model correctly classified 20 non-heart disease cases and 26 heart disease cases. However, it generated 8 false positive and 7 false negative predictions.

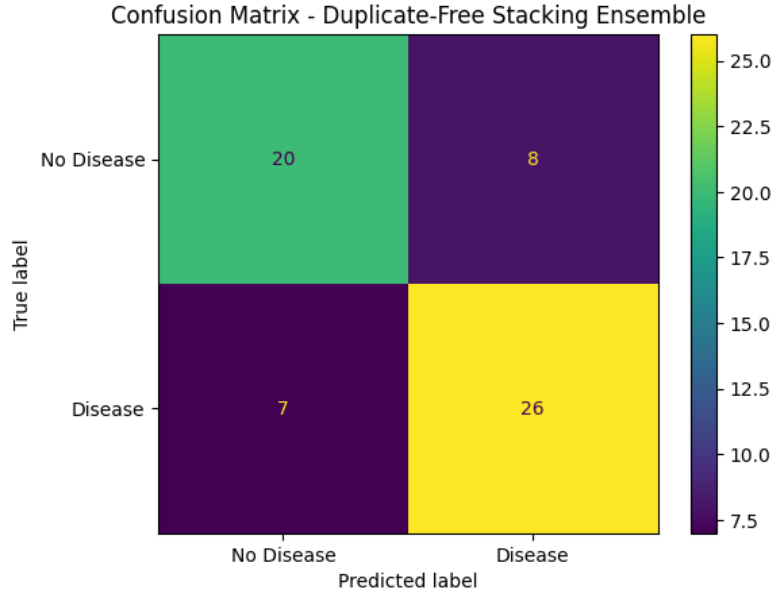


Figure 3: Confusion matrix of duplicate-free stacking ensemble.

The proposed optimised stacking ensemble demonstrated improved classification performance, as shown in Figure 4. The model correctly identified 21 non-heart disease cases and 29 heart disease cases. Compared with the baseline model, false negatives were reduced from 7 to 4 cases, while false positives decreased slightly from 8 to 7 cases.

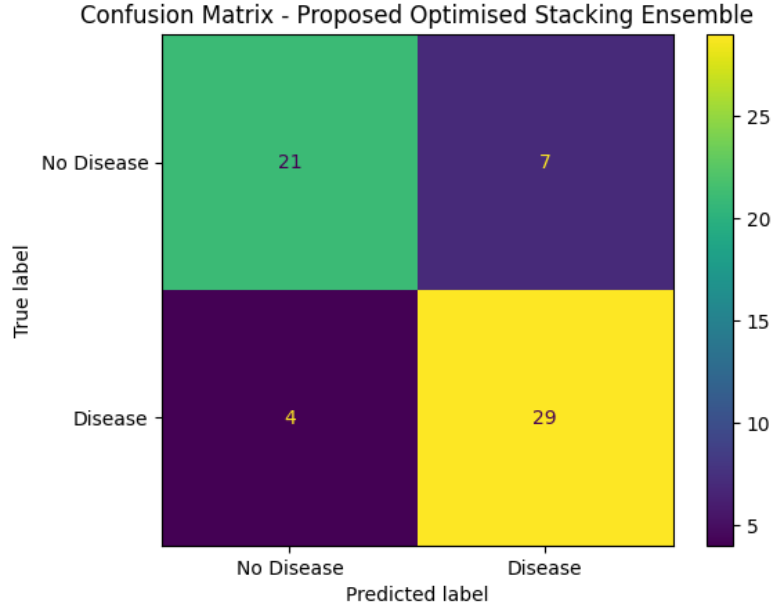


Figure 4: Confusion matrix of proposed optimised stacking ensemble.

The reduction in false negative predictions is particularly important for healthcare prediction tasks because false negatives represent patients with potential heart disease who are incorrectly classified as low risk. The proposed framework achieved a higher recall value of 87.88%, indicating an improved ability to identify positive heart disease cases.

Overall, confusion matrix analysis supports the quantitative performance improvements

observed in Section 6.2. The proposed optimised stacking ensemble not only improved overall metrics but also demonstrated better classification behaviour by reducing missed positive cases on the duplicate-free dataset.

6.4 ROC Curve Analysis

Receiver Operating Characteristic (ROC) curve analysis was performed to evaluate the discrimination capability of the proposed optimised stacking ensemble compared with the duplicate-free baseline model. The ROC curve illustrates the relationship between the true positive rate and false positive rate at different classification thresholds.

The area under the ROC curve (AUC) provides a single measure of model discrimination ability, where a higher value indicates better separation between heart disease and non-heart disease cases. In this study, ROC analysis was conducted using the testing predictions generated by both stacking approaches.

The duplicate-free baseline stacking ensemble achieved an AUC value of 0.868, while the proposed optimised stacking ensemble achieved an improved AUC value of 0.885. Figure 5 presents the ROC curve comparison between the two approaches.

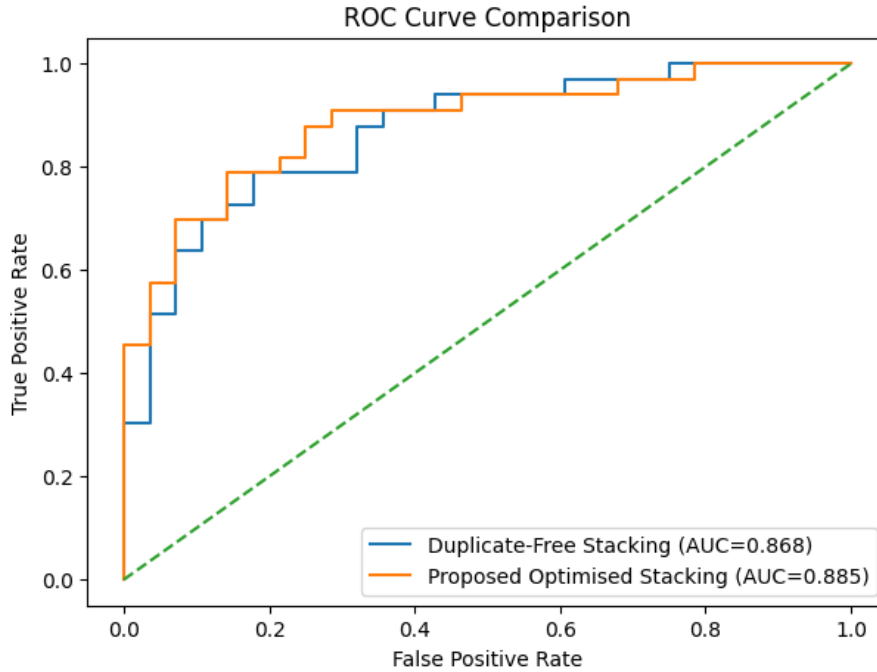


Figure 5: ROC curve comparison between duplicate-free and proposed optimised stacking ensembles.

The higher AUC value obtained by the proposed framework indicates improved discrimination capability after applying feature selection and hyperparameter optimisation. Although the improvement is relatively small, it demonstrates that the optimised model provides better classification performance while maintaining evaluation on unique patient records.

The ROC analysis further supports the findings from accuracy, F1-score, and confusion matrix evaluation. The proposed framework achieves a better balance between identifying

positive heart disease cases and limiting incorrect classifications, making it a more reliable approach compared with the duplicate-free baseline model.

6.5 SHAP Explainability Analysis

To improve the interpretability of the proposed optimised stacking ensemble, SHAP (SHapley Additive exPlanations) analysis was performed to investigate the contribution of individual clinical features to model predictions. While performance metrics demonstrate the predictive capability of a model, explainable analysis provides additional information regarding the factors influencing the classification decisions.

SHAP values quantify the contribution of each input feature to a prediction by measuring how the presence or absence of a feature affects the model output. The method provides both global explanations, which describe overall feature importance across the dataset, and local explanations, which describe the reasoning behind individual predictions [8].

In this study, SHAP analysis was applied to the proposed optimised stacking framework using the selected clinical features. The analysis identified the relative contribution of important variables including chest pain type (*cp*), maximum heart rate achieved (*thalach*), thalassemia indicator (*thal*), number of major vessels (*ca*), and oldpeak.

Figure 6 presents the SHAP summary plot, showing the distribution and direction of feature contributions across the testing samples.

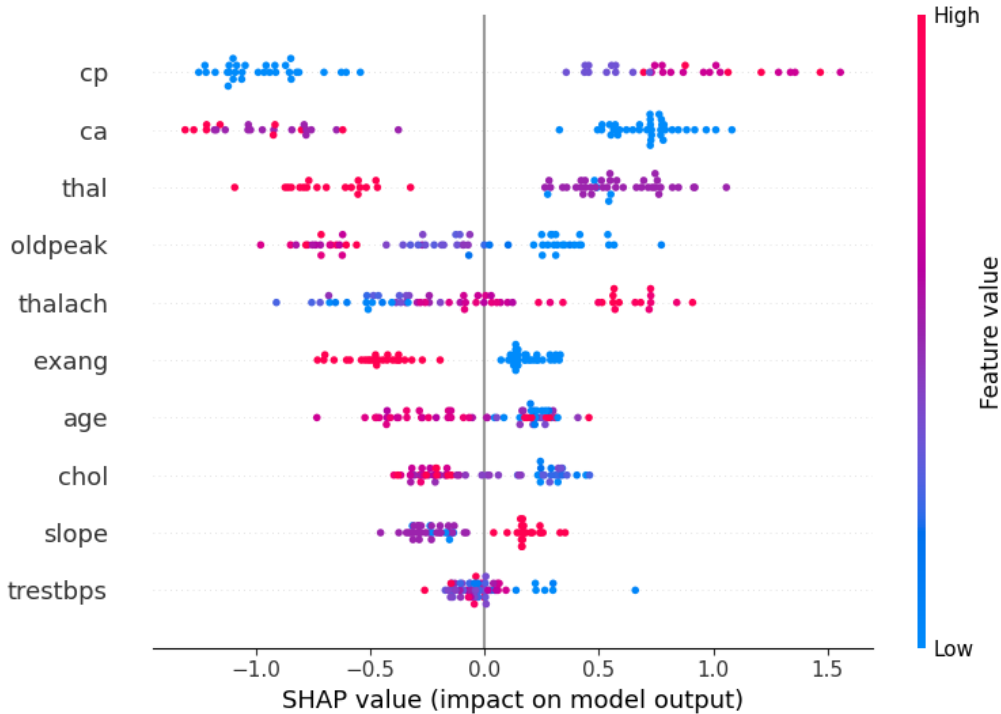


Figure 6: SHAP summary plot showing feature contributions in the proposed model.

The global SHAP feature importance ranking is presented in Figure 7. The results demonstrate that the selected clinical variables contribute differently to model predictions, with higher SHAP importance values indicating stronger influence on the classification outcome.

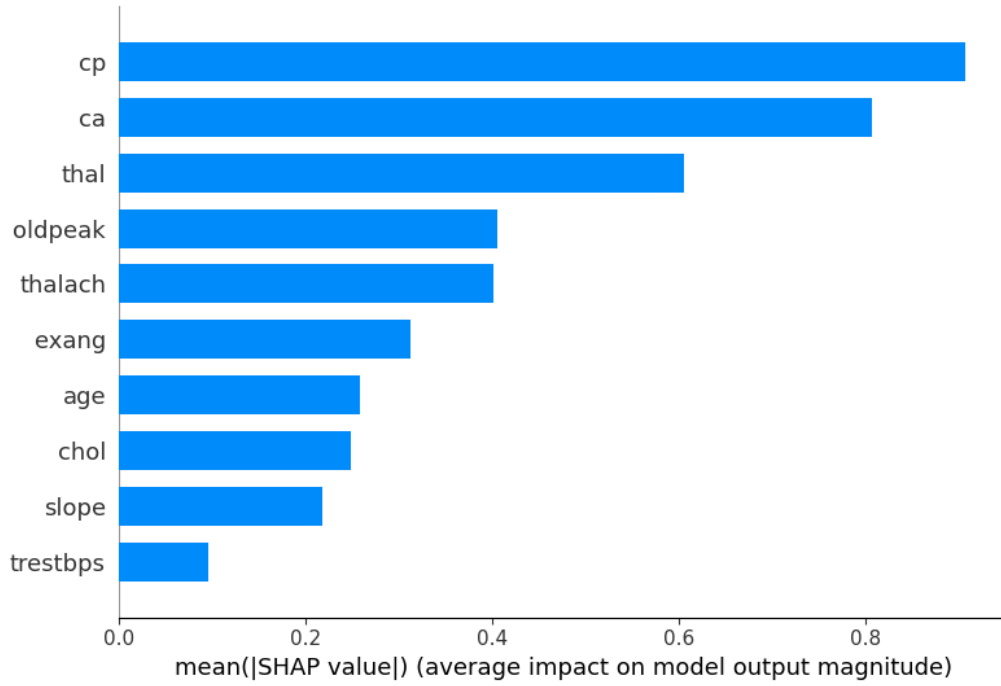


Figure 7: Global SHAP feature importance ranking of the proposed framework.

The explainability analysis provides additional confidence in the proposed framework by demonstrating that predictions are based on meaningful clinical attributes rather than only statistical patterns. Integrating SHAP with the optimised stacking model improves transparency and supports the development of more interpretable machine learning systems for healthcare prediction.

7 Discussion

7.1 Key Findings

This study investigated the reproducibility and improvement of a stacking-based machine learning framework for heart attack prediction. The research consisted of two major stages: reproduction of the existing stacking ensemble approach and development of an optimised explainable framework using a more reliable dataset evaluation strategy.

The reproduction experiment demonstrated that the original stacking methodology could be successfully implemented. Using the original dataset containing 1,025 records, the reproduced stacking ensemble achieved perfect classification performance across all evaluation metrics. However, further dataset analysis revealed a significant number of duplicated observations, suggesting that the reported performance may have been influenced by the characteristics of the dataset rather than representing true generalisation capability.

After removing duplicated records, the dataset was reduced to 302 unique observations. Evaluation on this duplicate-free dataset resulted in a more realistic performance level, demonstrating the importance of data validation in healthcare machine learning research. The decrease in performance after duplicate removal highlights that evaluation results should be interpreted carefully when datasets contain repeated samples.

The proposed optimised stacking framework improved performance compared with the duplicate-free baseline model. The integration of feature importance analysis, feature selection, and hyperparameter optimisation increased accuracy from 75.41% to 81.97% and improved F1-score from 77.61% to 84.06%. These results indicate that optimisation strategies can enhance predictive capability while maintaining evaluation on unique patient records.

Furthermore, SHAP-based explainability analysis provided additional insight into the prediction process by identifying the clinical variables contributing most to model decisions. This demonstrates that the proposed framework provides not only improved prediction performance but also greater transparency compared with a purely accuracy-focused approach.

Overall, the findings highlight that reliable healthcare machine learning systems require a combination of appropriate data validation, model optimisation, and interpretability techniques. The proposed framework addresses these requirements by balancing predictive performance with reliability and explainability.

7.2 Comparison with Original Approach

The proposed framework differs from the original stacking approach in terms of dataset evaluation strategy, feature optimisation, model tuning, and interpretability. While the original study focused primarily on achieving high predictive performance through a stacking ensemble architecture, this research further investigates the reliability and transparency of the prediction process.

The reproduced model achieved 100% accuracy when evaluated using the original dataset containing 1,025 records. This result was comparable to the high performance reported by Bhagat et al., who achieved 98.53% accuracy using their stacking-based framework [3]. However, subsequent duplicate analysis identified 723 repeated observations within the dataset, which

significantly affected the evaluation environment.

After removing duplicated records, the performance of the stacking ensemble decreased to 75.41% accuracy. This reduction indicates that the original performance evaluation was sensitive to dataset characteristics. Instead of optimising performance on the original dataset, the proposed framework focused on developing a more reliable model using unique patient records.

The proposed optimised stacking framework improved the duplicate-free baseline by incorporating feature importance analysis, feature selection, and hyperparameter optimisation. The final model achieved 81.97% accuracy, 84.06% F1-score, and 88.53% AUC on the duplicate-free dataset. Although these values are lower than the performance obtained from the original duplicated dataset, they provide a more realistic evaluation of the model’s ability to generalise to unseen patient records.

Another important difference is the integration of explainable artificial intelligence. The original stacking framework primarily focused on predictive performance, whereas the proposed approach incorporates SHAP analysis to identify the contribution of individual clinical features. This additional layer of interpretability is valuable in healthcare applications because understanding the reason behind predictions is important for user confidence and practical adoption.

Overall, the proposed framework does not aim to maximise accuracy under potentially biased evaluation conditions. Instead, it aims to achieve a balanced solution that considers predictive performance, dataset reliability, and model interpretability.

7.3 Practical Implications

The proposed optimised stacking ensemble framework demonstrates several practical implications for machine learning-based heart attack prediction systems. The results indicate that combining multiple classifiers with appropriate optimisation and explainability techniques can provide a more reliable approach for analysing clinical data.

One important implication of this research is the importance of data quality validation before applying machine learning models in healthcare. The identification of a large number of duplicated observations demonstrates that dataset characteristics can significantly influence reported model performance. Therefore, healthcare machine learning studies should include comprehensive data quality assessment as an essential part of the modelling process.

The proposed framework may support the development of decision-support systems by providing predictions together with explanations of important contributing features. The integration of SHAP analysis allows users to understand which clinical variables influence model decisions, improving transparency compared with models that provide only classification outputs.

The feature analysis results also provide insights into the clinical variables associated with model predictions. Important features identified by the framework, including chest pain characteristics, maximum heart rate, thalassemia indicators, and vessel-related measurements, may help researchers understand the factors considered by the prediction model.

However, the proposed framework should be considered as a research-oriented machine learning approach rather than a replacement for professional medical diagnosis. Further validation

using larger and more diverse clinical datasets, including external validation studies, would be required before considering practical healthcare implementation.

7.4 Limitations

Although the proposed framework achieved improved performance compared with the duplicate-free baseline model, several limitations should be considered when interpreting the findings of this study.

First, the evaluation was performed using a single heart disease dataset with a relatively limited number of unique observations. After duplicate removal, the dataset contained only 302 patient records. Although this cleaning process improved evaluation reliability, the reduced sample size may limit the ability of the model to capture the full diversity of real-world patient characteristics.

Second, the proposed framework was evaluated using an internal train-test split rather than external validation with an independent clinical dataset. Therefore, the generalisation capability of the model across different healthcare institutions, populations, and data collection environments remains uncertain.

Third, although SHAP analysis improved model interpretability, explainability methods provide insights into model behaviour rather than direct clinical causation. The identified important features represent variables used by the machine learning model for prediction and should not be interpreted as independent medical risk factors without further clinical investigation.

Finally, the proposed framework focuses on improving predictive reliability, optimisation, and transparency. Additional factors such as real-time clinical integration, patient privacy considerations, and large-scale deployment requirements were beyond the scope of this research.

Future studies should address these limitations by using larger multi-centre datasets, conducting external validation, and evaluating the framework under real-world healthcare conditions.

8 Future Work

Although the proposed optimised stacking ensemble framework improved the reliability and interpretability of heart attack prediction, several opportunities remain for further research and development.

First, future studies should evaluate the framework using larger and more diverse clinical datasets collected from multiple healthcare institutions. External validation on independent datasets would provide a stronger assessment of model generalisation capability and reduce dependency on a single dataset.

Second, future research can investigate advanced machine learning and deep learning approaches to determine whether more complex architectures can provide additional improvements while maintaining interpretability. Comparative studies between traditional ensemble methods and modern deep learning models could provide further insights into their suitability for healthcare prediction tasks.

Third, the proposed framework can be extended by exploring additional explainable artificial intelligence techniques beyond SHAP. Combining multiple explanation approaches may provide a more comprehensive understanding of model behaviour and improve transparency for different stakeholders.

Fourth, future work may investigate privacy-preserving machine learning approaches, such as federated learning, to enable collaborative model development across healthcare organisations while reducing the need to share sensitive patient data.

Finally, future research should explore practical deployment considerations, including integration with clinical decision-support systems, real-time prediction requirements, and continuous model monitoring. Such developments would require careful evaluation to ensure that machine learning systems remain reliable, transparent, and appropriate for healthcare environments.

9 Conclusion

This study investigated the reproducibility and improvement of a stacking-based machine learning framework for heart attack prediction. The research first reproduced an existing stacking ensemble approach using six machine learning classifiers, including Logistic Regression, Decision Tree, Random Forest, XGBoost, Naive Bayes, and K-Nearest Neighbours. The reproduced model achieved extremely high performance on the original dataset, consistent with the strong results reported in the selected research study.

However, further analysis identified a significant dataset quality issue, where a large proportion of observations were duplicated. The presence of repeated samples highlighted the importance of careful dataset validation in healthcare machine learning research, as evaluation results may be influenced by data composition and experimental conditions.

To address this limitation, an improved explainable stacking ensemble framework was proposed using a duplicate-free dataset. The proposed approach incorporated duplicate removal, Random Forest-based feature importance analysis, feature selection, hyperparameter optimisation, and SHAP-based explainability. Evaluation on the unique dataset demonstrated that the proposed framework improved prediction performance compared with the duplicate-free baseline stacking model, achieving 81.97% accuracy, 84.06% F1-score, and 88.53% AUC.

Beyond improving predictive performance, the proposed framework also enhanced model transparency through explainable artificial intelligence analysis. SHAP analysis provided insights into the contribution of important clinical features, allowing better understanding of the factors influencing model predictions.

Overall, this research demonstrates that developing reliable healthcare machine learning systems requires more than achieving high accuracy. Proper dataset validation, optimisation strategies, and interpretability techniques are essential for building trustworthy predictive models. The proposed framework provides a balanced approach by combining predictive capability with improved reliability and explainability for heart attack prediction applications.

Generative AI Acknowledgement

Generative Artificial Intelligence tools were used during this project for planning, brainstorming, debugging assistance, and language improvement. All technical decisions, implementation choices, experimental analysis, and final conclusions were reviewed and validated by the author.

10 Project Resources

The complete project materials, including source code, dataset, notebook, documentation, and supplementary files, are available through the following resources:

- **GitHub Repository:**

<https://github.com/nafishasnat2081/SIT720-Heart-Attack-Prediction-ML>

- **Presentation Video:**

<https://youtu.be/PWXu9QFd00E>

- **Google Drive Archive:**

Complete project archive containing all submission materials.

https://drive.google.com/drive/folders/1eHh1__wCu5-yzjuMpmKDsVacsm3jWwAa?usp=sharing

References

- [1] S. Mohan, C. Thirumalai, and G. Srivastava, “Effective heart disease prediction using hybrid machine learning techniques,” *IEEE Access*, vol. 7, pp. 81 542–81 554, 2019.
- [2] S. Uddin, A. Khan, M. E. Hossain, and M. A. Moni, “Comparing different supervised machine learning algorithms for disease prediction,” *BMC Medical Informatics and Decision Making*, vol. 19, p. 281, 2019.
- [3] M. Bhagat, D. Kumar, and A. Sharma, “An efficient stacking-based ensemble technique for early heart attack prediction,” *Multimedia Tools and Applications*, vol. 84, pp. 36 351–36 375, 2025.
- [4] L. Breiman, “Random forests,” *Machine Learning*, vol. 45, pp. 5–32, 2001.
- [5] T. Chen and C. Guestrin, “Xgboost: A scalable tree boosting system,” in *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 2016, pp. 785–794.
- [6] A. K. Garate-Escamila, A. Hassan, and E. Andr  s, “Classification models for heart disease prediction using feature selection and machine learning techniques,” *Computers in Biology and Medicine*, vol. 124, p. 103948, 2020.
- [7] M. Kumar *et al.*, “A comparative performance assessment of optimized multilevel ensemble learning model with existing classifier models,” *Big Data Mining and Analytics*, vol. 5, no. 3, pp. 371–387, 2022.
- [8] S. M. Lundberg and S.-I. Lee, “A unified approach to interpreting model predictions,” in *Advances in Neural Information Processing Systems*, vol. 30, 2017.
- [9] C. Molnar, “Interpretable machine learning: A guide for making black box models explainable,” 2022.
- [10] S. M. Lundberg, G. Erion, H. Chen *et al.*, “From local explanations to global understanding with explainable ai for trees,” *Nature Machine Intelligence*, vol. 2, pp. 56–67, 2020.